Volume 03 - Book 03.09 Swarm Intelligence

Version v11 draft

README.md

Book 03.09 - Swarm Intelligence

Version: v11 draft

Status: Draft

Document ID: MAL-V03-B03-09-README

Purpose

Define the Swarm Intelligence blueprint package and its subsystem decomposition as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for the Swarm Intelligence blueprint package and its subsystem decomposition. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for the Swarm Intelligence blueprint package and its subsystem decomposition.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Swarm Intelligence.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for the Swarm Intelligence blueprint package and its subsystem decomposition.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.

- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.
- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- SwarmIntelligenceDefined
- SwarmIntelligenceRegistered
- SwarmIntelligenceStateChanged
- SwarmIntelligenceReviewRequested
- SwarmIntelligenceGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.

- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.
- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Swarm Intelligence has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B009
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v11 draft	2026-06-29	Draft	Initial Swarm Intelligence blueprint content for Mariam Architecture Library v11.

Book 03.09 - Swarm Intelligence

Version: v11 draft

Status: Draft

Document ID: MAL-V03-B03-09-INDEX

Purpose

Define the Swarm Intelligence blueprint package and its subsystem decomposition as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for the Swarm Intelligence blueprint package and its subsystem decomposition. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for the Swarm Intelligence blueprint package and its subsystem decomposition.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Swarm Intelligence.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for the Swarm Intelligence blueprint package and its subsystem decomposition.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.
- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

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Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
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Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
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Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.

- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.
- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Swarm Intelligence has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B009
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v11 draft	2026-06-29	Draft	Initial Swarm Intelligence blueprint content for Mariam Architecture Library v11.

book-03-09-01-swarm-intelligence-overview.md

Book 03.09.01 - Swarm Intelligence Overview

Version: v11 draft

Status: Draft

Document ID: MAL-V03-B03-09-001

Purpose

Define Swarm Intelligence Overview within the Swarm Intelligence blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Swarm Intelligence Overview within the Swarm Intelligence blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Swarm Intelligence Overview within the Swarm Intelligence blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Swarm Intelligence.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Swarm Intelligence Overview within the Swarm Intelligence blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.
- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- SwarmIntelligenceOverviewDefined
- SwarmIntelligenceOverviewRegistered
- SwarmIntelligenceOverviewStateChanged
- SwarmIntelligenceOverviewReviewRequested
- SwarmIntelligenceOverviewGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.

- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.
- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Swarm Intelligence Overview has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B009
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v11 draft	2026-06-29	Draft	Initial Swarm Intelligence blueprint content for Mariam Architecture Library v11.

book-03-09-02-swarm-chief.md

Book 03.09.02 - Swarm Chief

Version: v11 draft

Status: Draft

Document ID: MAL-V03-B03-09-002

Purpose

Define Swarm Chief within the Swarm Intelligence blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Swarm Chief within the Swarm Intelligence blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Swarm Chief within the Swarm Intelligence blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Swarm Intelligence.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Swarm Chief within the Swarm Intelligence blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- SwarmChiefDefined
- SwarmChiefRegistered
- SwarmChiefStateChanged
- SwarmChiefReviewRequested
- SwarmChiefGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Swarm Chief has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B009
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v11 draft	2026-06-29	Draft	Initial Swarm Intelligence blueprint content for Mariam Architecture Library v11.

Book 03.09.03 - Coordinator

Version: v11 draft

Status: Draft

Document ID: MAL-V03-B03-09-003

Purpose

Define Coordinator within the Swarm Intelligence blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Coordinator within the Swarm Intelligence blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Coordinator within the Swarm Intelligence blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Swarm Intelligence.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Coordinator within the Swarm Intelligence blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- CoordinatorDefined
- CoordinatorRegistered
- CoordinatorStateChanged
- CoordinatorReviewRequested
- CoordinatorGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Coordinator has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B009
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v11 draft	2026-06-29	Draft	Initial Swarm Intelligence blueprint content for Mariam Architecture Library v11.

book-03-09-04-teams.md

Book 03.09.04 - Teams

Version: v11 draft

Status: Draft

Document ID: MAL-V03-B03-09-004

Purpose

Define Teams within the Swarm Intelligence blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Teams within the Swarm Intelligence blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Teams within the Swarm Intelligence blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Swarm Intelligence.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Teams within the Swarm Intelligence blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- TeamsDefined
- TeamsRegistered
- TeamsStateChanged
- TeamsReviewRequested
- TeamsGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Teams has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B009
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v11 draft	2026-06-29	Draft	Initial Swarm Intelligence blueprint content for Mariam Architecture Library v11.

Book 03.09.05 - Dynamic Swarms

Version: v11 draft

Status: Draft

Document ID: MAL-V03-B03-09-005

Purpose

Define Dynamic Swarms within the Swarm Intelligence blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Dynamic Swarms within the Swarm Intelligence blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Dynamic Swarms within the Swarm Intelligence blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Swarm Intelligence.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Dynamic Swarms within the Swarm Intelligence blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- DynamicSwarmsDefined
- DynamicSwarmsRegistered
- DynamicSwarmsStateChanged
- DynamicSwarmsReviewRequested
- DynamicSwarmsGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Dynamic Swarms has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B009
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v11 draft	2026-06-29	Draft	Initial Swarm Intelligence blueprint content for Mariam Architecture Library v11.

Book 03.09.06 - Parallel Execution

Version: v11 draft

Status: Draft

Document ID: MAL-V03-B03-09-006

Purpose

Define Parallel Execution within the Swarm Intelligence blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Parallel Execution within the Swarm Intelligence blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Parallel Execution within the Swarm Intelligence blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Swarm Intelligence.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Parallel Execution within the Swarm Intelligence blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- ParallelExecutionDefined
- ParallelExecutionRegistered
- ParallelExecutionStateChanged
- ParallelExecutionReviewRequested
- ParallelExecutionGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Parallel Execution has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B009
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v11 draft	2026-06-29	Draft	Initial Swarm Intelligence blueprint content for Mariam Architecture Library v11.

Book 03.09.07 - Recovery

Version: v11 draft

Status: Draft

Document ID: MAL-V03-B03-09-007

Purpose

Define Recovery within the Swarm Intelligence blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Recovery within the Swarm Intelligence blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Recovery within the Swarm Intelligence blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Swarm Intelligence.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Recovery within the Swarm Intelligence blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- RecoveryDefined
- RecoveryRegistered
- RecoveryStateChanged
- RecoveryReviewRequested
- RecoveryGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Recovery has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B009
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v11 draft	2026-06-29	Draft	Initial Swarm Intelligence blueprint content for Mariam Architecture Library v11.

Book 03.09.08 - Communication

Version: v11 draft

Status: Draft

Document ID: MAL-V03-B03-09-008

Purpose

Define Communication within the Swarm Intelligence blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Communication within the Swarm Intelligence blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Communication within the Swarm Intelligence blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Swarm Intelligence.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Communication within the Swarm Intelligence blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- CommunicationDefined
- CommunicationRegistered
- CommunicationStateChanged
- CommunicationReviewRequested
- CommunicationGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Communication has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B009
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v11 draft	2026-06-29	Draft	Initial Swarm Intelligence blueprint content for Mariam Architecture Library v11.

Book 03.09.09 - Optimization

Version: v11 draft

Status: Draft

Document ID: MAL-V03-B03-09-009

Purpose

Define Optimization within the Swarm Intelligence blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Optimization within the Swarm Intelligence blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Optimization within the Swarm Intelligence blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Swarm Intelligence.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Optimization within the Swarm Intelligence blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- OptimizationDefined
- OptimizationRegistered
- OptimizationStateChanged
- OptimizationReviewRequested
- OptimizationGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Optimization has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B009
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v11 draft	2026-06-29	Draft	Initial Swarm Intelligence blueprint content for Mariam Architecture Library v11.

Book 03.09.10 - Consensus

Version: v11 draft

Status: Draft

Document ID: MAL-V03-B03-09-010

Purpose

Define Consensus within the Swarm Intelligence blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Consensus within the Swarm Intelligence blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Consensus within the Swarm Intelligence blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Swarm Intelligence.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Consensus within the Swarm Intelligence blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- ConsensusDefined
- ConsensusRegistered
- ConsensusStateChanged
- ConsensusReviewRequested
- ConsensusGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Consensus has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B009
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v11 draft	2026-06-29	Draft	Initial Swarm Intelligence blueprint content for Mariam Architecture Library v11.

Book 03.09.11 - Critique Loops

Version: v11 draft

Status: Draft

Document ID: MAL-V03-B03-09-011

Purpose

Define Critique Loops within the Swarm Intelligence blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Critique Loops within the Swarm Intelligence blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Critique Loops within the Swarm Intelligence blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Swarm Intelligence.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Critique Loops within the Swarm Intelligence blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- CritiqueLoopsDefined
- CritiqueLoopsRegistered
- CritiqueLoopsStateChanged
- CritiqueLoopsReviewRequested
- CritiqueLoopsGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Critique Loops has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B009
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v11 draft	2026-06-29	Draft	Initial Swarm Intelligence blueprint content for Mariam Architecture Library v11.

Book 03.09.12 - Conflict Resolution

Version: v11 draft

Status: Draft

Document ID: MAL-V03-B03-09-012

Purpose

Define Conflict Resolution within the Swarm Intelligence blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Conflict Resolution within the Swarm Intelligence blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Conflict Resolution within the Swarm Intelligence blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Swarm Intelligence.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Conflict Resolution within the Swarm Intelligence blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- ConflictResolutionDefined
- ConflictResolutionRegistered
- ConflictResolutionStateChanged
- ConflictResolutionReviewRequested
- ConflictResolutionGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Conflict Resolution has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B009
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v11 draft	2026-06-29	Draft	Initial Swarm Intelligence blueprint content for Mariam Architecture Library v11.

Book 03.09.13 - Task Allocation

Version: v11 draft

Status: Draft

Document ID: MAL-V03-B03-09-013

Purpose

Define Task Allocation within the Swarm Intelligence blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Task Allocation within the Swarm Intelligence blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Task Allocation within the Swarm Intelligence blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Swarm Intelligence.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Task Allocation within the Swarm Intelligence blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- TaskAllocationDefined
- TaskAllocationRegistered
- TaskAllocationStateChanged
- TaskAllocationReviewRequested
- TaskAllocationGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Task Allocation has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B009
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v11 draft	2026-06-29	Draft	Initial Swarm Intelligence blueprint content for Mariam Architecture Library v11.

Book 03.09.14 - Swarm Memory

Version: v11 draft

Status: Draft

Document ID: MAL-V03-B03-09-014

Purpose

Define Swarm Memory within the Swarm Intelligence blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Swarm Memory within the Swarm Intelligence blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Swarm Memory within the Swarm Intelligence blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Swarm Intelligence.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Swarm Memory within the Swarm Intelligence blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- SwarmMemoryDefined
- SwarmMemoryRegistered
- SwarmMemoryStateChanged
- SwarmMemoryReviewRequested
- SwarmMemoryGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Swarm Memory has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B009
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v11 draft	2026-06-29	Draft	Initial Swarm Intelligence blueprint content for Mariam Architecture Library v11.

Book 03.09.15 - Swarm Governance

Version: v11 draft

Status: Draft

Document ID: MAL-V03-B03-09-015

Purpose

Define Swarm Governance within the Swarm Intelligence blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Swarm Governance within the Swarm Intelligence blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Swarm Governance within the Swarm Intelligence blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Swarm Intelligence.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Swarm Governance within the Swarm Intelligence blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- SwarmGovernanceDefined
- SwarmGovernanceRegistered
- SwarmGovernanceStateChanged
- SwarmGovernanceReviewRequested
- SwarmGovernanceGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Swarm Governance has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B009
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v11 draft	2026-06-29	Draft	Initial Swarm Intelligence blueprint content for Mariam Architecture Library v11.

Book 03.09.16 - Swarm Events

Version: v11 draft

Status: Draft

Document ID: MAL-V03-B03-09-016

Purpose

Define Swarm Events within the Swarm Intelligence blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Swarm Events within the Swarm Intelligence blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Swarm Events within the Swarm Intelligence blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Swarm Intelligence.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Swarm Events within the Swarm Intelligence blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- SwarmEventsDefined
- SwarmEventsRegistered
- SwarmEventsStateChanged
- SwarmEventsReviewRequested
- SwarmEventsGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Swarm Events has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B009
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v11 draft	2026-06-29	Draft	Initial Swarm Intelligence blueprint content for Mariam Architecture Library v11.

Book 03.09.17 - Swarm Storage

Version: v11 draft

Status: Draft

Document ID: MAL-V03-B03-09-017

Purpose

Define Swarm Storage within the Swarm Intelligence blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Swarm Storage within the Swarm Intelligence blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Swarm Storage within the Swarm Intelligence blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Swarm Intelligence.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Swarm Storage within the Swarm Intelligence blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- SwarmStorageDefined
- SwarmStorageRegistered
- SwarmStorageStateChanged
- SwarmStorageReviewRequested
- SwarmStorageGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Swarm Storage has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B009
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v11 draft	2026-06-29	Draft	Initial Swarm Intelligence blueprint content for Mariam Architecture Library v11.

book-03-09-18-swarm-security.md

Book 03.09.18 - Swarm Security

Version: v11 draft

Status: Draft

Document ID: MAL-V03-B03-09-018

Purpose

Define Swarm Security within the Swarm Intelligence blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Swarm Security within the Swarm Intelligence blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Swarm Security within the Swarm Intelligence blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Swarm Intelligence.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Swarm Security within the Swarm Intelligence blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- SwarmSecurityDefined
- SwarmSecurityRegistered
- SwarmSecurityStateChanged
- SwarmSecurityReviewRequested
- SwarmSecurityGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Swarm Security has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B009
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v11 draft	2026-06-29	Draft	Initial Swarm Intelligence blueprint content for Mariam Architecture Library v11.

book-03-09-19-swarm-testing.md

Book 03.09.19 - Swarm Testing

Version: v11 draft

Status: Draft

Document ID: MAL-V03-B03-09-019

Purpose

Define Swarm Testing within the Swarm Intelligence blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Swarm Testing within the Swarm Intelligence blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Swarm Testing within the Swarm Intelligence blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Swarm Intelligence.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Swarm Testing within the Swarm Intelligence blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- SwarmTestingDefined
- SwarmTestingRegistered
- SwarmTestingStateChanged
- SwarmTestingReviewRequested
- SwarmTestingGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
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Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Swarm Testing has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B009
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v11 draft	2026-06-29	Draft	Initial Swarm Intelligence blueprint content for Mariam Architecture Library v11.

book-03-09-20-swarm-roadmap.md

Book 03.09.20 - Swarm Roadmap

Version: v11 draft

Status: Draft

Document ID: MAL-V03-B03-09-020

Purpose

Define Swarm Roadmap within the Swarm Intelligence blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Swarm Roadmap within the Swarm Intelligence blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Swarm Roadmap within the Swarm Intelligence blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Swarm Intelligence.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Swarm Roadmap within the Swarm Intelligence blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- SwarmRoadmapDefined
- SwarmRoadmapRegistered
- SwarmRoadmapStateChanged
- SwarmRoadmapReviewRequested
- SwarmRoadmapGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
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Acceptance Criteria

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Version History

Version	Date	Status	Notes
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